Table 2 (reproduction, pilot, $B = 48$ per cell). Relative bias of exposure effect estimates of cumulative total exposure with model misspecification (bold cells with correct model specification).

	Relative bias (%)					
	Naive		RRC corrected			
$\rho(c, C)$	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M
A. Mean relative bias over converged fits; the paper's values in (gray)						
Cox+Additive Cox true $\beta = 1$, additive independent error						
0.3	−37.8 (−33.5)	6.7 (−28.3)	**−2.6 (−4.4)**	−22.8 (−5.7)	276.9 (5.9)	174.3 (6.6)
0.6	−30.0 (−32.8)	29.6 (−27.9)	**5.3 (−4.0)**	−5.2 (−5.5)	172.8 (5.6)	165.4 (6.1)
0.9	−28.6 (−31.4)	46.0 (−27.2)	**6.9 (−3.8)**	1.2 (−5.4)	367.2 (5.3)	23.5 (5.7)
Cox+Multiplicative Cox true $\beta = 1$, multiplicative independent error						
0.3	−29.2 (−34.7)	72.4 (−29.0)	12.9 (−6.1)	**6.7 (−5.7)**	214.3 (4.8)	173.6 (4.3)
0.6	−36.2 (−33.9)	42.3 (−28.4)	−5.8 (−5.6)	**−4.9 (−5.3)**	107.6 (4.5)	110.4 (4.1)
0.9	−32.1 (−32.6)	48.9 (−27.8)	1.8 (−5.4)	**1.8 (−5.0)**	124.1 (4.0)	123.3 (3.7)
ERR+Additive ERR true $\beta = 1.109$, additive independent error						
0.3	−77.2 (−42.1)	−67.8 (−35.8)	−66.9 (−14.9)	−72.8 (−15.2)	**−36.3 (−5.5)**	23.0 (−6.6)
0.6	−68.2 (−40.6)	−55.0 (−33.0)	−52.0 (−14.5)	−58.7 (−14.9)	**31.8 (−5.1)**	40.0 (−6.4)
0.9	−72.2 (−38.9)	−57.2 (−31.2)	−58.5 (−14.2)	−61.1 (−14.7)	**166.2 (−4.6)**	8.6 (−6.2)
ERR+Multiplicative ERR true $\beta = 1.109$, multiplicative independent error						
0.3	−52.2 (−43.0)	−9.0 (−36.2)	−24.5 (−16.8)	−27.9 (−16.0)	51.2 (−6.7)	**38.7 (−6.3)**
0.6	−54.4 (−41.7)	−12.1 (−33.4)	−31.0 (−15.8)	−31.7 (−15.6)	34.3 (−5.8)	**31.3 (−6.1)**
0.9	−61.0 (−40.4)	−25.4 (−31.9)	−41.6 (−14.7)	−41.2 (−15.2)	11.0 (−4.8)	**12.6 (−5.7)**
B. Median relative bias over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]						
Cox+Additive Cox true $\beta = 1$, additive independent error						
0.3	−37.7	−5.8 [2]	**−1.0**	−19.3	4408.6 [33]	483.4 [19]
0.6	−24.6	23.5 [3]	**12.1**	0.7	4408.6 [37]	4408.6 [30]
0.9	−21.7	47.1 [1]	**17.3**	14.9	4408.6 [35]	4408.6 [32]
Cox+Multiplicative Cox true $\beta = 1$, multiplicative independent error						
0.3	−28.9	53.2	10.0	**6.0**	140.2	135.9
0.6	−37.5	20.1	−5.3	**−7.4**	83.3	73.9
0.9	−30.3	35.3	3.8	**3.5**	98.0	103.1
ERR+Additive ERR true $\beta = 1.109$, additive independent error						
0.3	−74.0	−70.4	−60.5	−71.4	**−18.8 [14]**	−47.8 [7]
0.6	−62.2	−60.1	−44.6	−69.1	**98.4 [17]**	−47.1 [10]
0.9	−72.6	−68.6	−58.4	−64.5	**60.7 [15]**	−40.1 [11]
ERR+Multiplicative ERR true $\beta = 1.109$, multiplicative independent error						
0.3	−46.3	−17.6	−23.4	−24.4	24.2	**21.1**
0.6	−55.1	−23.2	−32.0	−31.9	24.5	**19.5**
0.9	−62.5	−40.9	−44.0	−43.5	−11.3	**−10.2**

Cox model $\lambda(t) = \lambda_0(t)\exp(\beta x(t))$ with true $\beta_{\text{Cox}} = 1$; ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{\text{ERR}} = 1.109$. Relative bias of each fit is against the true β of the model it fits: $100(\hat{\beta} - \beta)/\beta$.
$n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. Surrogate C has compound symmetry ($\rho = 0.5$) across years, $E(C) = 0.012$; true $c = \gamma_0 + \gamma_1 C + \epsilon$ (additive) or $\log c = \gamma_0 + \log C + \epsilon$ (multiplicative), $E(c) = 0.008$, with σ_ϵ set by $\rho(c, C)$; median $x \approx 0.18$.
– : no fit converged. A median of 4408.6 means at least half of the fits stopped at the cap $\beta = 50$.